Order Cetacea

The cetaceans are divided into two groups, Odontoceti (toothed cetaceans: dolphins, porpoises) and Mysticeti (baleen whales). The birth of a singleton, highly precocial offspring is normal for this order. Males have abdominal testes. Females have elongate mammary glands whose teats are enclosed within deep pockets on each side of the urogenital slit.

Several periodicals cover cetacean reproduction, among them are Aquatic Mammals, Cetology, Hvalradets Skrifter, Investigations on Cetacea, Marine Mammal Science, Norsk Hvalfangst-tidende (no longer in publication), Reports of the International Whaling Commission, and Scientific Reports of the Whales Research Institute.

Order Cetacea, Family Platanistidae

The 10-240 kg river dolphins (*Inia*, *Lipotes*, *Platanista*, *Pontoporia*) inhabit large river systems of Asia and South America, where they feed on a variety of vertebrates and invertebrates. Female *Pontoporia* are larger than males (5560) and ovulate primarily from their left ovary (4422). The villous, diffuse, epitheliochorial placenta implants into a bicornuate uterus which is asymmetrical in *Lipotes* (1778, 3739, 4418, 7563). In one observed captive birth, the young was born tail first (*Inia* 10949). Females have 2 U-shaped mammary glands (*Lipotes* 1778) and may have a postpartum estrus (*Pontoporia* 4423). Details of anatomy (1778, 4418, 4422, 4423), copulation (8529), and birth (5069, 10949) are available.

NEONATAL MASS		
Inia geoffrensis	6.8 kg; n1	10949
Platanista minor	7 kg	9175
NEONATAL SIZE		
Inia geoffrensis	76-80 cm	4421
	81 cm; n1	10949
	min 81 cm; n1	873
Lipotes vexillifer	est 80 cm	9049a
Platanista minor	45 cm	9175
Pontoporia blainvillei	est 59 cm; n1	4421
	TL: 63-70 cm; n5 term emb	1377
	70-85 cm	4423, 5560
LITTER SIZE		
Inia geoffrensis	1 yg	4421
	1 yg; n1 lt	10949
Lipotes vexillifer	1 yg	9049a
Platanista gangetica	1 emb; n1 f	8532
Platanista minor	1; sometimes 2	8530
Pontoporia blainvillei	1 yg	4421, 4423
SEXUAL MATURITY		
Inia geoffrensis	f: ovulate: 183 cm	873
	m: spermatogenesis: 198.5 cm	873
Lipotes vexillifer	f: ovulate: 8 yr, 184.5 cm	9049a
Platanista spp	~10 yr, 170 cm	5555
Platanista gangetica	f: 1st cl: 200 cm	4422
	m: testes: large 180-185 cm	4422
Pontoporia blainvillei	f: 1st cl: 2.7 yr; r2-4 yr, r137-146 cm	5560
	f: 1st cl: 151-170 cm	4422
	f: 137-145 cm	1377
	m: testes: large 2-3 yr, 131 cm, 25-29 kg	5560
GESTATION		
Inia geoffrensis	8.5 mo; n1	10949
Lipotes vexillifer	est 10-11 mo	9049a
Platanista minor	8-9 mo	9175
	12 mo; hunter reports	9175
	1 yr	8530
Pontoporia blainvillei	10.5 mo	5560
	est <11 mo; r10.5-12	4423
LACTATION		
Pontoporia blainvillei	wean: 9 mo	4423
	wean: ~1 yr; milk in stomach	5560

INTERLITTER INTERVAL

Lipotes vexillifer	est 2 yr	9049a
Pontoporia blainvillei	2 yr	5560
	2 yr	4423

SEASONALITY

Inia geoffrensis	birth: May-July; Amazon R, S America	873
	birth: July-Sept; upper Amazon R, S America	4421
Lipotes vexillifer	birth: Feb; Changjiang R, China	9049a
	birth: Feb-Mar; few data; Yangtze R, China	1778
	lact f: Dec; n1; China	1379
Platanista spp	birth: yr rd, small peaks Oct-Mar, June; India	5555
Platanista minor	mate: Mar; local fisherman; Pakistan	9175
	mate/birth: Mar; Indus R, Pakistan	8530
Pontoporia blainvillei	mate: Dec-Feb, peak early-mid Jan; La Plata, S America	4423
	birth: mid Nov-Dec; La Plata, S America	4423

Order Cetacea, Family Delphinidae

Marine dolphins (*Cephalorhynchus* 25-75 kg; *Delphinus* 40-160 kg; *Feresa* 160-170 kg; *Globicephala* 800-2900 kg; *Grampus* 230-450 kg; *Lagenodelphis* 130-210 kg; *Lagenorhynchus* 30-235 kg *Lissodelphis* 80-115 kg; *Orcaella* 125-200 kg; *Orcinus* 2200-9000 kg; *Peponocephala* 180-205 kg; *Pseudorca* 370-1700 kg; *Sotalia* 30-80 kg; *Sousa* 140-285 kg; *Stenella* 35-165 kg; *Steno* 100-145 kg; *Tursiops* 115-650 kg) are social, aquatic carnivores distributed throughout the world's oceans and some tropical rivers. Males are often larger than females (393, 1975, 7123, 7370, 8022, 9784, 9787), but not in *Cephalorhynchus* (6539a, 10065a).

Both induced and spontaneous ovulation are reported for *Delphinus* and *Tursiops* (induced: 4420, 5085, 9407, 11980; spontaneous: 766, 4413, 4420, 4422, 5784, 8390). The left ovary may be dominant to the right one (1975, 4419, 4421, 5086, 6841, 7123, 7396, 8387, 8389, 9300; cf 4430, 8064). Vaginal plugs are reported (3093, 11980). The 0.5-26 kg, epitheliochorial, villous, diffuse placenta often implants in the left horn of the bicornuate or bipartite uterus (192, 765, 766, 1678, 1998, 4146, 4413, 4845, 7563, 10061, 10956, 11858). A postpartum estrus is suggested for *Delphinus* and *Stenella* (192, 5564, 8387, 8389, 10158, 11980). Neonates are precocious (5146, 7012, 10691). Testes and epididymides can reach 3% of body mass (*Cephalorhynchus* 10065a).

Details of anatomy (1975, 1976a, 1998, 4036, 4413, 4420, 4422, 4430, 4823, 5085, 6841, 9653, 9777), intersex (7926), endocrinology (4845, 5459, 5784, 9108, 9109, 9629, 9653, 9654, 9655, 11405, 11586), sperm (3309), copulation (3093, 6704, 8780, 11320), birth (431, 3092a, 3094, 5385, 7012, 10691), milk composition (8319, 8398, 8543, 11980), fetal growth rate (8390a), and reproductive energetics (817, 1767a, 10277) are available.

NEONATAL MASS

Cephalorhynchus commersonii	m: 4.5-5.5 kg; n1	5385
	m: 7.3 kg; n1 term emb	6539a
Cephalorhynchus heavisidii	m: 9.5 kg; n1 presumed newborn	869a
Delphinus delphis	6.8 kg; n1	6544
	7.35 kg; r6.8-7.9; n2	192
Globicephala melaena	180 kg; r165-190	9777
Lagenorhynchus acutus	24 kg	9789
Orcinus orca	est 140 kg; n1	431
	155 kg stillborn; 158 kg 11 d; est 160 kg; n3	765
	~180 kg	6945a
Stenella attenuata	>10 kg	7400
Stenella coeruleoalba	>10 kg	7400
Tursiops truncatus	11.3-15.9 kg	4022
	m: 12 kg; n1	9407
	f: 24.5 kg; r19-30; n2 stillborn	5567
	26.5 kg; r23-30; n2 3-4 d; 25 kg; 10 d; 30 kg; 32 d	5567
	m: 29 kg; n1 stillborn	5567

NEONATAL SIZE

Cephalorhynchus commersonii	m: 55-65 cm; n1	5385
	m: 73.4 cm; n1 term emb	6539a
Cephalorhynchus heavisidii	m: 84.7 cm; n1 presumed newborn	869a
Delphinus delphis	TL: 75-85 cm	10158
	75-90 cm	4420
	est 79.0-81.3 cm	8390a
	80-90 cm	8390a
	84 cm	8066
	86.4 cm; n1	6544
	90 cm	1975
	91.5 cm; r86-97; n2	192
	est 105 cm	9695a
Globicephala spp	160 cm; n1 smallest yg	4156a
Globicephala macrorhynchus	135-146 cm	12021a
	139.5 cm	5562
	140 cm	7370
Globicephala melaena	est 140 cm	8390a
	TL: ~160-170 cm	9776
	174 cm	8066
	f: 174 cm	9777
	175 cm	10313b
	m: 178 cm	9777
	m: 187 cm; n1	4135
Grampus griseus	est 110-120	8390a
	150 cm	7370
	150 cm	8390a
Lagenodelphis hosei	~100 cm	7983
Lagenorhynchus acutus	95 cm	7370
	~110-120 cm	9789
Lagenorhynchus albirostris	~122 cm	7983
	125 cm	8390a
	est 125 cm	9695a
Lagenorhynchus obliquidens	est 95-110 cm	4420
Orcinus orca	208-220 cm	8390a
	210 cm	5441
	210-250 cm	8390a
	TL: 210-270 cm	10158
	236.2 cm; n4	927a
	240 cm	7370
	246.4 cm; n1	927a
	270 cm	7929
Peponocephala electra	75-90 cm	2363
Pseudorca crassidens	est 150-210 cm	10292a
	est 160 cm	8390a
	193 cm	8797
Sotalia fluviatilis	71-83 cm	7983
	est ~75 cm	873
Sousa chinensis	est 100 cm	8390a
Stenella attenuata	est 82-85 cm; r71-92; n609 term emb, yg	4909
	82.5 cm; n86 emb, yg	8387
	89 cm	5564
Stenella coeruleoalba	est 99.8 cm	5554
	100 cm	7370
	100 cm; n68 emb, 20 yg	7396, 7399
	105 cm	8066
Stenella longirostris	est 76.5 cm	8388
	est 76.9 cm; from emb & sm yg	8389
	~80 cm	10158
Tursiops truncatus	81-122 cm	4022
	r83.8-112.0 cm; n8	9300
	95-115 cm	9653
	100 cm	9787
	110.3 cm; r98-118; n6	4420, 4422, 4430

	m: 112 cm; n1	9407
	115 cm	4908a
	r115-140 cm; n12	5567
	116 cm; r95-132; n9 f	4908a
	118 cm; r110-134; n9 m	4908a
	128 cm; n20, 0-11 d	5567
	est 130 cm	9695a
	135 cm	8066
WEANING MASS		
Delphinus delphis	40 kg	192
WEANING SIZE		
Delphinus delphis	150 cm	192
Orcinus orca	430 cm	4715
Stenella attenuata	f: r115-130 cm	8387
Stenella coeruleoalba	174 cm	7397
Tursiops truncatus	~150 cm; n1, 10 mo	9300
	195 cm; n1, 18 mo	9300
LITTER SIZE		
Cephalorhynchus commersonii	1; n1 lt	5385
	1 emb; n1 f	6539a
Cephalorhynchus eutropia	1 emb; n1 f	3890b
Cephalorhynchus heavisidii	1 emb; n1 f	869a
Cephalorhynchus hectori	1 yg	10065a
Delphinus delphis	1	192
	1 cl	10061
Feresa attenuata	1	8064
Globicephala macrorhynchus	1 emb ea; n141 f	5562
Globicephala melaena	1 emb ea; n2 f	7099
	1.7 ova/cycle	9777
Grampus griseus	1 emb; n1 f	9301a
Lagenodelphis hosei	1	7983
Lagenorhynchus acutus	1-2; 1 set of twins	5442
Lagenorhynchus albirostris	1 yg	8390a
Lagenorhynchus obliquidens	1 yg	4420
Orcaella brevirostris	1 yg ea; n2 lt	10678
Orcinus orca	1 yg	431
	1 yg	6945a
	1 yg; n5	765
Pseudorca crassidens	1 emb; n1 f	10158
	1 emb ea; n4 f	9696
Sotalia fluviatilis	1 emb; n1	5111
Sousa chinensis	1	8390a
Stenella attenuata	1.02 emb; r1-2; n59 f	5564
Stenella coeruleoalba	1	7396
	1.8 cl/cycle	7397
	2 emb; n1 f	10853
	2; n1 f	10853
Stenella longirostris	1 emb; 1 f	7401
Tursiops truncatus	1 cl	11858
	1 cl/yr	9787
	1 yg	7012
	1 yg ea; n3 f	244
	1 emb ea; n6 f	4156
	1 ea; n18 lt	12073
SEXUAL MATURITY		
Cephalorhynchus commersonii	f: ovaries mature: ~5 yr; ~165 cm	1976a
	f: ovaries mature: 5-6 yr; ~130 cm	3890a
	m: testicular sperm: 5-6 yr; 127-131 cm	3890a, 6539a
	m: testes mature: ~8 yr; ~165 cm	1976a
Cephalorhynchus hectori	f: 1st birth: est 7-9 yr	10065a
	m: testes large: 6-9 yr	10065a
Delphinus delphis	f: 1st cl: 7 yr	5085
	f: 1st cl: 165-182 cm	5086
	f: 1st cl: 195-196 cm; 6 layers dentine	1975, 1976

	f: 3 yr	4420
	f: 3 yr	10061
	f: 3.5 yr	6544
	f: TL 210-220 cm	10158
	m: spermatogenesis: 187-198 cm	4420
	m: testes large: 175-190 cm	5086
	m: testes large: >190 cm; 12 dentine layers	5085
	m: testes large: 200 cm; 5-7 layers dentine	1975, 1977
	m: 4 yr	10061
	m: 4.5 yr	6544
Globicephala macrorhynchus	f: 1st ovulate: 7-12 yr	6841
	f: 1st cl: 300-320 cm	7370
	f: sperm in cloaca: 8.25 yr	5562
	f: 9 yr	5563
	f: 380-390 cm	10158
	m: epididymal sperm: min 394 cm	5562
	m: 480 cm	7370
Globicephala melaena	f: 6-8 yr; 366 cm	9777, 9784
	m: 12 yr; 487 cm	9777, 9784
Grampus griseus	m: 300+ cm	7370
Lagenodelphis hosei	f: 1st preg: 236 cm; n1	10158
	f/m: ~7 yr; 230 cm	11096
Lagenorhynchus acutus	f: 1st cl: 201-222 cm	9789
	m: testes large: 230-240 cm; from graph	9789
Lagenorhynchus obliquidens	f: 1st cl: 173 cm; n1	4420, 4422
	f: 175-186 cm	7370
	m: testes large: >180 cm	4422
	m: sperm present: 200 cm; n1	4420
	m: 170-180 cm	7370
Lissodelphis borealis	f: 1st cl, 1st preg: 200-201 cm	6323
	m: spermatogenesis: 212-219.5 cm	6323
Orcinus orca	f: 1st cl: 480 cm	5441
	f: 1st preg: min 6.7 yr; 490 cm	927a
	f: 1st preg: 460 cm	1826
	f: 455-485 cm	6945a
	f: 460-540 cm	4715
	f: 490 cm	1375
	m: testes large: 570 cm	5441
	m: min 12 yr; 580-600 cm	927a
	m: 520-620 cm	4715
	m: 670 cm	1375
Peponocephala electra	f: <12 yr	7983
	f: >225 cm	1398
	m: <7 yr	7983
Pseudorca crassidens	f: 1st cl: est 358-440 cm	8022
	f: 1st cl: 395-427 cm	3482
	f: 1st cl: <427 cm	1998
	f: 1st preg: 360-420 cm	9784
	f: 8-14 yr; 366-427 cm	8797
	m: 8-14 yr; 366-427 cm	8797
	m: 511 cm	9784
	f/m: 8-10 yr	2045
Sotalia fluviatilis	f: 1st cl: 138.5 cm	873
	m: testes large: 139 cm	873
Stenella spp	f: 1st cl: 165-170 cm	4420
	m: sperm: 180-214 cm	4420
Stenella attenuata	f: 1st cl: >170 cm; n1	4422
	f: 1st conceive: 4.5 yr; r4.1-7.6; 181 cm; 50-60 kg	8387
	f: 1st cl: median est 9 yr	5556
	m: sperm: 180-214 cm; n6	4420, 4422
	m: testes large: 6 yr; 195 cm; 75 kg	8387
	m: testes large: median est 11.8 yr	5556
Stenella coeruleoalba	f: 5-6 yr; 210 cm	7370
	f: est 7.4-9.7 yr	5558

	f: 8.7-8.8 yr; 219 cm; 8.9 yr; 50% mature	7396, 7397, 7399
	f: 9 yr; 212 cm	5554, 5556
	f: *est* TL 210 cm	10158
	m: testes large: 8.8 yr, 50% mature	7399
	m: socially mature: 17 yr	7399
	m: 6.7-7.1 yr	7396
	m: 7-11 yr	5556
	m: 9.0 yr; 217 cm	7397
	m: 9 yr; 220 cm	5554
	f/m: 220 cm	4824
Stenella longirostris	f: 1st cl: 7-10 yr; 180 cm; 55 kg	7123
	f: est 3.7-5 yr; 165 cm	8389
	m: testes large: 192 cm; 65 kg	7123
	m: *est* 6-11.5 yr; 170 cm	8389
	f/m: ~5-6 yr; 164.1-167.2 cm	8388
	f/m: 170 cm	4422
Steno bredanensis	max 225 cm	7398
Tursiops truncatus	f: adult hormone levels: 5-7 yr	9653
	f: 1st cl: 12 yr; 235 cm	9787
	f: 1st cl: 202-231 cm; possibly 6-9 yr	9300
	f: 1st preg: 11-13 yr; 227-238 cm	1767b
	f: 1st birth: 5 yr; n1	7012
	f: ~5 yr; >220 cm	4420, 4430
	f: 6 yr	9175
	f: ~160-222.5 cm	4424
	m: spermatogenesis: ~9-11 yr	9300
	m: testes large: 13 yr; 245 cm	9787
	m: 1st sire: 10 or 11, 13 yr; n2	1767b
	m: 6-7 yr	4420
	m: ~10 yr; 260 cm	4430
	m: ~202 cm	4424
CYCLE LENGTH		
Stenella attenuata	1 mo	5564
Stenella coeruleoalba	4 mo	7396
GESTATION		
Delphinus delphis	5-6 mo	10061
	276 d	4422
	10-11 mo	192
	10-11 mo; n3 sources	8390a
	11 mo	1975
	11 mo	4420
	~11 mo	11980
Globicephala macrorhynchus	452 d; r402-512	5562
Globicephala melaena	15.5-16 mo	9777
	16 mo	3488
Grampus griseus	*est* ~1 yr	10158
Lagenorhynchus acutus	10 mo; n2 sources	8390a
	~10 mo	7370
	~11 mo	9789
Lagenorhynchus obliquidens	~10 mo	4146
Orcaella brevirostris	*est* 14 mo	10678
Orcinus orca	<1 yr	7929
	12 mo; n3 sources	8390a
	~15 mo; n1	2218
	est 15-16 mo	6945a
	~517 d; n1	431
Peponocephala electra	*est* ~1 yr	1398
	~1 yr	10158
Pseudorca crassidens	~13 mo	2045
	462 d; fetal growth data	8797
	15.5-15.7 mo	10292a
Sotalia fluviatilis	10.2 mo; fetal growth data	873
Stenella attenuata	11.2 mo; fetal growth data	5564
	11.5 mo; n6243 f; fetal growth data	8387, 8390

Stenella coeruleoalba	~12 mo; fetal growth data	5554, 7397
	13.4 mo; fetal growth data	7399
Stenella longirostris	est 10.6 mo; from brain size	8388, 8389
Tursiops truncatus	>11 <13 mo	7012
	11-12 mo	9175
	~1 yr	9300
	12 mo	244
	12 mo	9407
	12 mo	4430
	~12 mo	11980
	~12 mo	10691
	12 mo; r11.5-12.5	9653
	12-13 mo	5146
LACTATION		
Cephalorhynchus hectori	solid food: 6 mo	10065b
Delphinus delphis	solid food: 6.5 mo	192
	wean: 10 mo	192
	wean: 10 mo	1975
	wean: est 14 mo	8390a
	wean: est 19 mo; n4 sources	8390a
Globicephala macrorhynchus	wean: min 2 yr	5562
Globicephala melaena	wean: 20 mo	10158
	wean: 22 mo	9777
Lagenorhynchus acutus	wean: ~18 mo	9789
Orcinus orca	solid food: est <2 mo; 260 cm; n1	4714
	solid food: 5 mo; n1	4714
	solid food: 11 wk	431
	nurse: 12 mo	4715
	nurse: ~18 mo	431
	independent: 2 yr	4715
Pseudorca crassidens	wean: 6 mo	2045
	wean: est 18 mo	8390a
Sousa chinensis	wean: 4-6 mo	9175
	young accompany mother: min 6 mo	9175
Stenella attenuata	solid food: 0.25-0.5 yr	5564
	wean: est 11.2 mo; r9-12	8387, 8390
	wean: est 24.5 mo; demographic statistics	5564
Stenella coeruleoalba	solid food: 3 mo	7396
	solid food: 0.5 yr	7397
	wean: 6-12 mo	10158
	wean: 15 mo	2045
	wean: 1.4 yr	7399
	wean: 1.5 yr; demographic statistics	5554
	wean: min 18 mo	7370
	wean: 1.5 yr; r1-3	7396, 7397
Stenella longirostris	wean: 13.8 mo; r10.1-17.5; n2	8389
	wean: 15-19 mo; n3 sources	8390a
Tursiops truncatus	solid food: 6 mo	9300
	solid food: 6-7 mo	9653
	wean: 1 yr	10698
	wean: 18 mo	9653
	wean: ~18 mo	10691
	wean: max 1.5 yr	7012
	wean: ~2 yr	9175
	nurse: 29 mo	9300
INTERLITTER INTERVAL		
Cephalorhynchus hectori	min 2 yr	10065b
Delphinus delphis	1 yr	10061
	~2 yr	1975
Globicephala macrorhynchus	3 yr	10158
Globicephala melaena	3-4 yr	2045
	40 mo	9777
Orcinus orca	3 yr	927a
Pseudorca crassidens	3 yr	2045

Stenella attenuata	r25.4-26.2 mo, 3 different yr	8387, 8390
	38.6-57.5 mo	5564
Stenella coeruleoalba	3 yr	5554
	3 yr	7397
	4.3 yr	7399
Stenella longirostris	2.11-2.22 yr	8389
Tursiops truncatus	1 yr; n1	9300
	2 yr	7012
	~2 yr	6322
	2-4 yr	11889
	22.7-27.6 mo; 2 samples	2051
	4 yr; n1 f	9407
SEASONALITY		
Cephalorhynchus commersonii	birth: Jan-Mar; Argentina	3890a, 6539a
Cephalorhynchus eutropia	preg f: Mar; n1; Chile	3890b
Cephalorhynchus heavisidii	preg f: Apr; n1; near Walvis Bay, S Africa	869a
Cephalorhynchus hectori	birth: Nov-Feb; New Zealand	10065b
Delphinus delphis	spermatogenesis: Jan-July; English Channel	1975
	spermatogenesis: Feb-Mar, low Aug; Bay of Biscayne	1977
	spermatogenesis: Mar-Aug; Gulf of Gascogne	1975
	testes: active Dec-June/July; e N Atlantic	1977
	mate: Feb-May; captive, St Augustine, FL	3093
	mate: June-Oct, peak July-Aug; Black Sea	10061
	mate: not seasonal; San Diego, CA	5085
	birth: peak May/June-end summer; France	1975
	birth: Dec-Mar; CA	4420, 4422
	birth: spring-fall, peak June-July; Black Sea	10061
	birth: yr rd, peak summer; S Africa	10158
	birth: yr rd, peaks Feb-May, Sept-Nov; CA, FL	4430
	birth: mid-winter; s CA	7960
	young seen: July-Jan, peak Nov; Napier Hawk's Bay, New Zealand	192
Globicephala macrorhynchus	conceive: peak Apr-May; Japan	5562
	fetal length varied Oct; w North Atlantic	7370
	preg f: high June, low Dec; Japan	5562
	birth: peak July-Aug; Japan	5562
	birth: Aug-winter; CA, Mexico	7960
Globicephala melaena	mate: Apr-May; e NEW, Canada	9777
	birth: ?-Oct, peak Aug; e NEW, Canada	9777
Grampus griseus	birth: est Dec-Apr; S Africa	10158
Lagenorhynchus acutus	birth: May-Aug, peak June/July; ME, MA	9789
Lagenorhynchus obliquidens	birth: May-Sept; s CA	4420
	varied length of young: summer-fall; CA, Mexico	7960
Lagenorhynchus obscurus	mate: late summer; S Africa	10158
Orcinus orca	mate: May-July; Japan	7929
	mate: Nov-Jan; n Atlantic	10158
	mate: May-July; n Pacific	10158
	conceive: peak Sept-Jan; Norway	1826
Peponocephala electra	preg f: Aug; e Australia	2363
	birth: Aug-Dec, s hemisphere	1398
Sotalia fluviatilis	conceive: Dec-Jan; Amazon, S America	873
	birth: Oct-Nov; Amazon, S America	873
Sousa spp	birth: yr rd, 50/75 Dec-Feb; S Africa	9408
Stenella spp	birth: Apr-June; Mexico	4420
Stenella attenuata	mate: Feb-Mar, July, Nov; Japan	5564
	birth: spring, fall; Carribean, Atlantic	4422
	birth: peak spring, fall, some summer; tropical Pacific	8387
	birth: yr rd, peak Feb-Apr; north popn also peak Sept; Pacific	594
Stenella coeruleoalba	testes: peak Oct, min Feb; Japan	7399
	conceive: Jan, July; Japan	7399
	mate: Jan-Feb, May-June, Sept-Oct; Japan	7397
	mate: Feb-May, July-Sept, Dec; Japan	7396
	birth: yr rd, peaks Dec-Feb, July-Aug; Japan	7396

Stenella longirostris	testosterone: high Apr-July; captive	11586
	birth: Mar-Apr, Sept; Pacific	594
	birth: mid May-mid June; FL	7123
	birth: yr rd; Pacific	594
Tursiops truncatus	mate: yr rd, peak Sept-Dec; captive	9407, 10693
	birth: Feb-May, Aug-Dec; n33; captive	11889
	birth: Mar-May; Pakistan	9175
	birth: Feb-June; England, UK	4420
	birth: yr rd, peak Feb-May, Sept-Nov; captive	4420, 4430
	birth: yr rd, peak Mar-Apr, Oct; FL	2051
	birth: est May; CA; few data; captive	2051
	birth: yr rd, peak Oct-Mar; captive, S Africa	9407
	birth: yr rd; few data; S Africa	9300
	birth: yr rd; n Gulf of Mexico	6322
	birth: yr rd; n18; captive	12073

Order Cetacea, Family Phocoenidae

Porpoises (*Neophocaina* 25-40 kg; *Phocoena* 25-90 kg; *Phocoenoides* 80-150 kg) are piscivores in temperate and arctic waters of both hemispheres. Females are larger than males (*Phocoena* 11191). Ovulation is usually from the left ovary (1777, 3276, 7506, 10063; cf 8064). A 700 g (n1), epitheliochorial, diffuse or discoidal placenta implants into the left horn of the bicornuate or bipartite uterus (1777, 2287, 4146, 4428, 5238, 5843, 7506, 7563, 10063, 10737, 11849). Details of anatomy (1777, 4428, 7506, 7507) and milk composition (1034, 11698a, 11980) are available.

NEONATAL MASS		
Neophocaena phocaenoides	m: 6.6 kg; n1 stillborn	5567
	m: 7.2, 7.4 kg; n2; 4, 17 d	5567
Phocoena phocoena	m: 3.1 kg; n1 stillborn	5238
	5-8 kg	5517
	m: 9.1 kg; n1 near term	3276
	f: 9.5 kg; n1 near term	3276
NEONATAL SIZE		
Neophocaena phocaenoides	77 cm	2365
Phocoena dioptrica	188.5 cm; r186-191; n2 term emb	5278
Phocoena phocoena	m: 40.3 cm; n1 stillbom	5238
	60 cm; n1 near term	5843
	67-80 cm	11191
	70-99 cm	9983
	70.5 cm; r70-71; n2	3626
	75 cm	8066
	76.2-99.1 cm	3626
	m: 78.5 cm; r70-87 mm; n2 near term	3276
	est 80 cm	9695a
	80-86 cm	3626
	82-85 cm	12058
	f: 84 cm; n1 near term	3276
Phocoena sinus	m: 74 cm; n1 yg with fetal folds	1378
Phocoena spinipinnis	44 cm; n1 near term	1380
Phocoenoides dalli	95-100 cm	5279
	100 cm	5557
WEANING MASS		
Neophocaena phocaenoides	24.5 kg; n1 immature	4428
WEANING SIZE		
Neophocaena phocaenoides	130 cm; n1 immature	4428
LITTER SIZE		
Neophocaena phocaenoides	1 emb; n1 f	5567
Phocoena dioptrica	1 emb ea; n2 f	5278
Phocoena phocoena	1 emb; n1 f	5238
	1 emb; n1 f	5843
	1 emb ea; n2 f	3276

Phocoena sinus	1 yg; n1 f	1378
Phocoena spinipinnis	1 emb; n1 f	7370
Phocoenoides dalli	1 emb ea; n39 f	5557
SEXUAL MATURITY		
Neophocaena phocaenoides	f: 1st cl: 141 cm; n1	4428
	m: spermatogenesis: 140-150 cm	4428
Phocoena phocoena	f: 1st preg: 14 mo, ~50 kg	7422
	f: ~4 yr, 145 cm	3276
	m: ~3 yr, 133 cm	3276
	m: 6 yr	11191
	f/m: 3-4 yr	3626
Phocoenoides dalli	f: 3 yr, 170-171 cm	5279
	f: 6.8 yr, 186.5 cm	5557
	f: 7 yr	11191
	f: 170 cm	7370
	m: 5-6 yr, ~182 cm	5279
	m: 7.9 yr, 195.7 cm	5557
	m: 185 cm	7370
	f/m: min 2 yr	5566
GESTATION		
Neophocaena phocaenoides	10 mo	1777
	~11 mo	5561
Phocoena spp	10 mo	4146
Phocoena phocoena	10-11 mo	7422
	~11 mo	3276
	332 d	6279
Phocoenoides dalli	7-9 mo	5279
	<1 yr	7370
	~12 mo; field data	5557
LACTATION		
Neophocaena phocaenoides	wean: 6-15 mo	5561
Phocoena phocoena	wean: 8 mo	7422
Phocoenoides dalli	wean: 1.6-4 mo	5279
	wean: 3 mo	5279
	wean: 2 yr	5557
INTERLITTER INTERVAL		
Neophocaena phocaenoides	mode 2 yr; r1-2	5561
Phocoena phocoena	1 yr	12058
	1 yr	7422
Phocoenoides dalli	3 yr	5557
SEASONALITY		
Neophocaena phocaenoides	mate: est May-Sept; Japan	5561
	mate: est June-July; Yangtse R, China	1777
	birth: est Apr; Yangtse R, China	1777
	birth: Apr-Aug, est peak Apr-May; Japan	5561
	f with young: peak Oct-Nov; few data; Pakistan	9175
Phocoena dioptrica	preg f: July-Aug; n2; se S America	5278
Phocoena phocoena	testes: large May-June, decline mid-Aug; w Atlantic	3276
	spermatogenesis: early phases May-July, peak mid-end July; w Atlantic	3276
	mate: end June/July-early Aug; w Atlantic	3276
	mate: July-Oct; Russia	12058
	conception: peak Aug; Denmark, Netherlands	6279
	birth: Apr-mid June; Russia	12058
	birth: est May-June; n Pacific	9983
	birth: June-July; w Atlantic	3276
Phocoena spinipinnis	preg f: Feb; n1; Uruguay	1380
Phocoenoides dalli	mate: mid Aug-Oct; Japan	5557
	birth: end July-early Aug; CA	7370
	birth: Aug-Sept; Japan	5557
	birth: est yr rd; few data; Pacific	7506

Order Cetacea, Family Monodontidae

Belugas (*Delphinapterus leucas*) and narwhals (*Monodon monoceros*) are 1000-2000 kg, gregarious carnivores of arctic waters. The long, spiral tusk commonly protruding from the upper jaw of male, and occasionally female, narwhals may be used in male-male competition (9977). Males are usually larger than females (2643, 6790, 9786, 11312). Ovulation may be induced (11980). The left ovary is slightly dominant (8064). The epitheliochorial placenta of *Delphinapterus* has branched chorionic villi (11980) and usually implants into the left uterine horn (10393). The uterus is bicornuate (*Delphinapterus* 11980) or bipartite (*Monodon* 358). Concurrent gestation and lactation as well as accessory corpora lutea have been reported for *Delphinapterus* (1303, 9781, 11980). Details of anatomy (358) and milk composition (6245, 11980) are available.

NEONATAL MASS		
Delphinapterus leucas	35-85 kg	10393
	57 kg	2643
Monodon monoceros	80+ kg	6790
NEONATAL SIZE		
Delphinapterus leucas	150 cm	6279
	150 cm	8066
	150 cm; r120-183; term emb, small calves	9781
Monodon monoceros	153 cm; n1 smallest calf	9006
	~160 cm	6790
	165 cm; n1 term emb	9006
LITTER SIZE		
Delphinapterus leucas	1	9781
	1.02; r1-2; n49 lt	2643
	1.06; r1-2; n17 lt	11312
Monodon monoceros	1 yg	6790
	mode 1 yg; r1-2	9006
SEXUAL MATURITY		
Delphinapterus leucas	f: 1st cl: 5 yr	9781
	f: 1st cl: 3.45 m	11312
	f: 1st preg: 5 yr; r4-7	10393
	f: 1st preg: 3.15-3.20 m	11312
	f: 3 yr, 270 cm	2643
	f: 5 yr	7370
	m: testicular histology: 7-9 yr	9781
	m: 4 yr	11312
	m: 8-9 yr	7370
	f/m: 2-5 yr	11980
	f/m: 5 yr	1302
GESTATION		
Delphinapterus leucas	11.5 mo	11980
	12 mo?	11312
	est 14 mo	9781
	14 mo	7370
	14 mo; fetal growth data	9781
	14 mo	1474
	14.5 mo	1302
	14-15 mo	10393
Monodon monoceros	est ~14 mo; fetal growth data	874
	est 15 mo	6790
LACTATION		
Delphinapterus leucas	solid food: 1 yr	10393
	wean: 20-24 mo	10393
	wean: est 21 mo; field data	9781
	wean: 2 yr	1302
Monodon monoceros	wean: est 20 mo; field data	6790
INTERLITTER INTERVAL		
Delphinapterus leucas	2-3 yr	7370
	3 yr	9781
Monodon monoceros	2-3 yr	874

SEASONALITY		
Delphinapterus leucas	mate: end Feb-June, est peak Mar; AK	10393
	mate: May; Cumberland Sound, Baffin I, NWT, Canada	1302
	mate: May-June; Gulf of St Lawrence	11312
	mate: June-Sept; Hudson Bay	10393
	preg f: July; Gulf of St Lawrence	11312
	birth: peak end Mar; w Greenland	10393
	birth: Mar-May; Arctic, n Atlantic, Hudson Bay	1474
	birth: May-June; Gulf of St Lawrence	6279, 11312
	birth: June-Aug, peak July; Gulf of St Lawrence	9785
	birth: peak end June-July, some Sept; w Hudson Bay	9781
	birth: July; Chukcki Sea, Mackensie, Bering Sea	10393
	birth: end July-early Aug; Cumberland Sound, Baffin I, NWT, Canada	1302
Monodon monoceros	mate: mid Apr; Canada, nw Greenland	6790
	birth: end June-Aug; arctic	874
	birth: mid July; Canada, nw Greenland	6790

Order Cetacea, Family Physeteridae

Sperm whales (*Kogia* 150-400 kg; *Physeter* 12,000-48,000 kg) are worldwide, oceanic predators primarily of squid. Pygmy sperm-whale males and females are of equal size, while males of *Physeter macrocephalus* are much larger than females (183, 393, 871, 2045). Mating occurs with ventral-ventral orientation (871). The placenta is epitheliochorial (11980). Females ovulate late in lactation (861, 3591, 11980). Details of anatomy (860, 861, 862, 863, 3590, 6961), sperm counts (1007), birth (8399), and milk composition (871, 5293, 11980, 12090) are available.

NEONATAL MASS		
Kogia breviceps	82 kg	7498c
Physeter macrocephalus	774 kg	10158
	1054 kg	3590
NEONATAL SIZE		
Kogia breviceps	113.5 cm; n1 large emb	9301
	116.5 cm; r110-123; n2 large emb	2766
	124.8 cm; r120-129.5; n2 small yg	1546, 1546a
	TL: 163 cm	7498c
Kogia simus	est 100 cm	9301
Physeter macrocephalus	360-430 cm	1474, 2045
	m: 382 cm; se12; n6	871
	390 cm	8066
	392 cm	1887
	f: 393 cm; se10; n9	871
	395-425 cm	5146
	~400 cm	6961
	TL: ~400 cm	10158
	404 cm	861
	est 405 cm	8065
	est 415 cm	9695a
	425-455 cm	7403
WEANING MASS		
Kogia breviceps	62.6 kg; n1 small m with squid in stomach	10078a
WEANING SIZE		
Kogia breviceps	TL: 155 cm; n1 small m with squid in stomach	10078a
Physeter macrocephalus	670 cm	8065
	< 760 cm	861
LITTER SIZE		
Kogia breviceps	1 emb ea; n2 f	2766
	1 emb; n1 f	9301
Kogia simus	1 yg	9301

Physeter macrocephalus	1.00 emb; twins 0.40%; n503 f	871
	1.00 emb; twins 0.45%; n2664 f	8065
	1.01 cl; 0.67% 2, 0.22% 3; n446 f	8065
	1.02 emb; twins 1.3%, triplets 0.01%; n748 f	5498
	1.17 emb; r1-2; n6 f	5559
SEXUAL MATURITY		
Kogia breviceps	f: 1st preg: est 2.7-2.8 m	9301
	m: testes enlarge: est 2.7-3.0 m	9301
Kogia simus	f: 1st preg: est 2.1-2.2 m	9301
	m: 1st sperm: est 2.0-2.2 m	9301
Physeter macrocephalus	f: 1st cl: 15 mo	6961
	f: 1st cl: 50% 9 yr, 8.5 m	861, 864
	f: 1st cl: 9.2 yr	8065
	f: 1st cl: 8.4-8.6 m	1888
	f: 1st cl: 8.7 m	1887
	f: 50% mature: 12-13 yr; 8.8 m; n256	564
	f: 1st preg: 9.0-11.1 m	7403
	f: 1st preg/lact: 9.75-10.05 m; n5	5559
	f: 54 mo	393
	f: 8.8 m	1375
	f: 9.15 m	8110
	m: spermatogenesis: 2 yr	6961
	m: spermatogenesis: 9.6-9.8 m	1888
	m: sperm: est 10.5-11 m	7927, 7931, 7932, 7934
	m: sperm: 11.27 m; ±0.44; n162	6527
	m: sperm: 12.5 m	871
	m: testes large: 9.33 m; ±0.51; n162	6527
	m: testes large: 12.5 m	7374
	m: 50% mature: 11 yr; 8.5-8.8 m; n781	564
	m: puberty: 10.7-11.6 m; sexual maturity: 25 yr, 13.72 m	7371
	m: puberty: 11.9-12.2 m; sexual maturity: 13.7-14.0 m	862, 864
	m: 9.5-10 m	1007
	m: 11.9 m	1375
	m: 14.20 m	393
	f/m: 9-10 yr	2045
	f/m: <9.15 m	7403
GESTATION		
Kogia simus	questionable est ~9 mo	7744
Physeter macrocephalus	14-15 mo; field data	871
	14.6 mo; fetal growth data	861
	14.75 mo	3591
	15-16 mo; n1706; fetal growth data	3488
	473 d; n166; fetal growth data	3588
	479 d; n55; fetal growth data	564
	16 mo	393
	16 mo; fetal growth data	1887
	~16 mo	6961
	16.4 mo; field data	8065
	17 mo	7403
	518 d; n277; fetal growth data	1888
LACTATION		
Kogia breviceps	solid food: 1.6 m; n1	1546
Physeter macrocephalus	solid food: ~1 yr, 2 dental layers	871
	nurse: est 5-6 mo or 17-18 mo	1888
	wean: 6 mo	5146
	wean: ~6 mo	2045
	wean: > 6 mo	6961
	wean: >1 yr	10698
	wean: ~13 mo	1887
	wean: 1.6-2.4 yr; f <20 yr; 2.2-4.0 yr; f >20 yr	871
	wean: 24 mo	393
	wean: 24-25 mo	861
	wean: 24-25 mo	3590, 3591

INTERLITTER INTERVAL		
Physeter macrocephalus	~2 yr	6961
	3 yr	564
	3-4 yr	8065
	4 yr	3590
	4 yr	861
SEASONALITY		
Kogia breviceps	birth: Feb; n1; e US	183
Physeter macrocephalus	ovarian activity: low June, peak Aug east, Oct west; S Africa	871
	spermatogenesis: yr rd; w S Africa	862
	mate: Mar; Japan	7403
	mate: Mar-May; n Hemisphere	3488
	mate: Apr-May; WA, Australia	7373
	mate: peak Apr-May; Azores	7373
	mate: Apr-July; CA	7373
	mate: Apr-Aug; ne Pacific	9091
	mate: Apr-Feb, most June-Dec, peak Sept; se Pacific	1888
	mate: peak Aug-Dec, most Oct; S Africa	6961
	mate: Sept; s Pacific	7373
	mate: Sept-Dec; s Hemisphere	3488
	mate: est Oct-Dec; S Africa	871
	mate: Oct-Apr, peak Dec; S Africa	3588
	mate: peak Dec-Jan; WA, Australia	564
	mate: yr rd, peak Apr; Japan	8065
	ovulate: end Aug-Mar, peak Dec; w S Africa	861
	conceive: Feb-June, older f earlier; Japan	8065
	conceive: Sept-Mar, 80% Nov; S Africa	3590, 3591
	conceive: end Aug-Mar, peak early Dec; w S Africa	861
	birth: Jan-Apr, peak Feb-Mar; S Africa	871
	birth: end Jan-mid Aug, peak Apr; S Africa	3588
	birth: peak Feb; s Pacific	7373
	birth: Apr-May; WA, Australia	564
	birth: May-Dec, peak July-Dec; Azores	1887
	birth: end July-Oct; Japan	7403
	birth: Aug-Sept; Japan	3726
	birth: Aug-Dec; ne Pacific	9091
	birth: peak Aug-Dec; CA	7373
	birth: Spet-July, most Nov-May, peak Feb; se Pacific	1888
	birth: Nov-June, peak Feb-Mar; w S Africa	861
	birth: peak Dec-Jan; S Georgia, s Atlantic Ocean	7373
	birth: yr rd, peak Jan-Mar; S Africa	6961

Order Cetacea, Family Ziphiidae

The relatively unstudied, beaked whales (*Berardius* 9000 kg; *Hyperoodon* 3500-5000 kg; *Indopacetus* est 2200 kg; *Mesoplodon* 1300-1500 kg; *Tasmacetus* 2500 kg; *Ziphius* 2450-2950 kg) are carnivores from both the Atlantic and the Pacific in arctic, temperate, and tropical waters. *Indopacetus* is known only from two skulls and *Tasmacetus* from a dozen specimens. *Berardius* has an epitheliochorial placenta (11980). The presence of old and new corporea lutea in the same female (*Berardius*, *Ziphius*), as well as the capture of a concurrently lactating and pregnant female (*Hyperoodon*), suggests a postpartum estrus (8059, 11980). Details of anatomy (2438, 7370) and milk composition (11031, 11980) are available.

NEONATAL SIZE		
Berardius bairdii	460 cm	8066
Hyperoodon ampullatus	305 cm	771
	350 cm; n1 term emb	8059
Mesoplodon bidens	150-240 cm	5440
Mesoplodon carlhubbsi	250 cm	7124
Ziphius cavirostris	200-305 cm	7370
	TL: 228-240 cm	10158
WEANING SIZE		
Mesoplodon bidens	315 cm; n1; est recently weaned	5440

LITTER SIZE		
Berardius bairdii	1 emb; n1 f	9091
Hyperoodon ampullatus	1	8059
Mesoplodon bidens	1 yg	5440
Mesoplodon carlhubbsi	1 emb; n1 f	7124
Mesoplodon mirus	1 emb; n1 f	1291
Ziphius cavirostris	1	10158
SEXUAL MATURITY		
Berardius bairdii	f: 10 m	6953
Hyperoodon ampullatus	f: 1st cl: 8-13 yr	773
	f: 1st cl: ~9 yr; 7.2-7.3 m	1824
	f: 1st preg: 6.60-6.80 m	771
	f: 6.7-7.01 m	7370
	m: spermatogenesis: 7-9 yr	773
	m: testes histology: 9-11 yr, 7.20 m	771
	m: 7.3 m	7370
Mesoplodon densirostris	f: 9 yr	7983
Ziphius cavirostris	f: 1st cl?: 6.1 m	7370
	m: testes size?: 5.4 m	7370
GESTATION		
Berardius arnuxii	~10 mo; few data	7014
Berardius bairdii	10 mo; few data	7014
	est 17 mo; fetal growth	5556a
Hyperoodon ampullatus	est 12 mo	8059
	~12 mo	771
Mesoplodon bidens	est 1 yr	5440
Mesoplodon carlhubbsi	12 mo	7983
LACTATION		
Hyperoodon ampullatus	wean: 1 yr; n1; squid and milk in stomach	773
Mesoplodon bidens	wean: est 1 yr	5440
LITTERS/YEAR		
Hyperoodon ampullatus	1	8059
SEASONALITY		
Berardius arnuxii	birth: end spring-early summer; New Zealand	7014
Hyperoodon ampullatus	mate: peak Apr; n Atlantic	771
	mate: Apr-May; Arctic	8059
	birth: peak Apr; n Atlantic	771
	birth: end Apr-June; Arctic Sea	8059
Hyperoodon planifrons	birth: spring-early summer; S Africa	10158
Mesoplodon bidens	birth: end winter-spring; Norway	5440
Mesoplodon carlhubbsi	birth: est June-Aug; n4; CA	7124
Mesoplodon densirostris	birth: summer; S Africa	10158
Mesoplodon stejnegeri	lact f: Oct; n1; AK	11031
Ziphius cavirostris	birth: summer; S Africa	10158

Order Cetacea, Family Eschrichtidae, Genus *Eschrichtius*

The migratory, grey whale (*Eschrichtius robustus*) is a 20,000 to 37,000 kg, baleen whale that inhabits in temperate and arctic Pacific waters and feeds on krill. Females are reportedly larger than males (302). The non-deciduate, epitheliochorial, diffuse placenta implants into a bipartite uterus (9094). Conception may occur during lactation (9094, 11979). Details of anatomy (9094), copulation (532, 9506), and milk composition (11979, 11980, 12090) are available.

NEONATAL MASS		
Eschrichtius robustus	f: 409 kg; n1 term emb	9094
	500 kg	11880
	680 kg	3788
NEONATAL SIZE		
Eschrichtius robustus	360-570 cm	302
	366-518 cm	9154
	425-515 cm	5146
	446.7 cm; r354-540; n69	8755

	450 cm	8066
	m: 456 cm; n24 term emb	9094
	458 cm; sd49; r350-630; n133 term emb, yg	10548
	468.3 cm; r395.5-540.5; n6	2887
	f: 469 cm; n17 term emb	9094
	~560 cm	3788
WEANING SIZE		
Eschrichtius robustus	850 cm	9094
LITTER SIZE		
Eschrichtius robustus	1 yg	9094
SEXUAL MATURITY		
Eschrichtius robustus	f: 1st cl: 8 yr; r5-11, 11.7 m	9094, 11880
	f: 1st cl: 7-11 yr	9018
	f: 1st cl, emb: 8-12 yr, ~12 m	11979
	f: 1st cl: 11.7-11.8 m	1008, 1009
	f: 1st preg: 12.7 m; n1	9091
	m: spermatogenesis: median 8 yr; r5-11	9018
	m: testes large: 6-8 yr, 11 m	11979
	m: 5-11 yr, 11.1 m	9094
	m: ~11.5 m	1008
	m: 11.9 m	9091
	f/m: 4-5 yr	3788
CYCLE LENGTH		
Eschrichtius robustus	40 d	9094, 11880
GESTATION		
Eschrichtius robustus	11-12 mo	3789
	est ~1 yr	5054
	~1 yr	302
	~1 yr	11979
	400 d; field data	9094, 11880
	est 418 d	8755
LACTATION		
Eschrichtius robustus	wean: 137 d	11979
	wean: 5 mo	5146
	wean: 6 mo	10548
	wean: 7 mo	9018
	wean: 7 mo	9094, 11880
INTERLITTER INTERVAL		
Eschrichtius robustus	1-2 yr	11979
	2 yr?	3789
	2 yr	9094
SEASONALITY		
Eschrichtius robustus	spermatogenesis: peak Nov-Jan; CA	11880
	estrus: end Nov-early Dec; CA	9094, 11880
	mate: Jan-Apr; CA	532
	mate: June; AK, Siberia	9506
	mate: Dec; Korea	302
	mate: Dec-May; e Pacific	5054
	mate: Dec; e Pacific	8755
	preg f: Jan; n14 near term; Mexico	9154
	birth: Jan-Feb; CA, Mexico	3789
	birth: Jan-Feb; ne Pacific	8755
	birth: Nov-Dec; Korea	302
	birth: Dec-Feb; Mexico	5054
	birth: end Dec-early Feb, peak Jan; CA	9094, 11880
	birth: winter; AK	6726
	wean: July-Aug; CA, Chukchi Sea	1009

Order Cetacea, Family Balaenopteridae

Rorquals (*Balaenoptera* 8000-172000 kg; *Megaptera* 25000-48000 kg) are aquatic, baleen whales with a worldwide distribution. Females are larger than males (1801, 2045, 3593). The diffuse, villous placenta implants into a bicornuate or bipartite uterus (691, 6958, 7563, 8108, 10526, 10969, 11361). Ovulation may be spontaneous (6836). A postpartum estrus occurs except in *Balaenoptera acutorostrata* (869, 1802, 3591, 5157, 6252, 6281). Details of anatomy (224, 691, 1799, 1800, 2480, 3038, 3589, 6702, 6958, 6960, 9186, 11178), endocrinology (392, 8743), sperm counts (1007), intersex (563), and milk composition (859, 869, 1802, 2439, 4056, 5295, 6245, 6702, 8072, 8073, 8321, 10611, 11634, 11980, 12090) are available.

NEONATAL MASS		
Balaenoptera borealis	650 kg	6279
Balaenoptera musculus	2500 kg	6279
Balaenoptera physalus	1900 kg	6279
Megaptera novaeangliae	1300 kg	6279
NEONATAL SIZE		
Balaenoptera acutorostrata	224 cm; n1	7099
	240-270 cm	6952
	250 cm	8066
	250-300 cm	4147
	est 270 cm	9695a
	280 cm	5157
	280 cm	5182
Balaenoptera borealis	TL: 360 cm	10158
	est 400 cm	4147
	420-450 cm	7403
	440 cm	6279
	440 cm	8066
	450 cm	2045
	450 cm	6700
	est 450 cm	9695a
	800-900 cm	6960
Balaenoptera edeni	TL: 340 cm	10158
	396 cm	866
Balaenoptera musculus	700 cm	6279
	700 cm	6700, 6702
	700 cm	8066
	est 700 cm	9695a
	750-800 cm	4147
Balaenoptera physalus	630-660 cm	7403
	640 cm	6279
	640 cm; r625-670	8063, 8066, 8071
	640-670 cm	7403
	650 cm	6700, 6702
	est 650 cm	9695a
Megaptera novaeangliae	410 cm; mode 420; r240-540; n240 term emb	1802
	430 cm	8066
	450-500 cm	6700
	450-500 cm	6958
	456 cm	6279
	est 460 cm	9695a
WEANING MASS		
Balaenoptera musculus	17,000 kg; 6 mo; from graph	3592
WEANING SIZE		
Balaenoptera borealis	TL: 8.5 m	10158
Balaenoptera musculus	TL: 16 m; 6 mo; from graph	3592
	16 m	5630
Balaenoptera physalus	est 12 m	8063
	12 m	5630
	~12.6 m	6702
Megaptera novaeangliae	7.5-8.0 m	6958
	TL: 8-9 m	10158

LITTER SIZE

Balaenoptera acutorostrata	1 cl	10969
	1.01 yg; r1-2; 0.65% twins; n1998 lt	8069
	Siamese twin emb; n1 f	12109
Balaenoptera borealis	1.02 yg; r1-2; n1098 lt	5730
	1.07 cl; r1-2; n27 f	3589
Balaenoptera edeni	1 emb ea; n2 f	8101
Balaenoptera musculus	1 emb; n~50 f	6702
	1.01 emb; 0.67% twins; n593 f; 7 emb; n1 f	9154
	1.01 emb; 0.74% twins; n272 f	1294
	1.01 emb; 0.78% twins; n19057 f	5730
	1.01 yg; r1-2; 0.87% twins; n19005 lt	5730
	2; n1 lt	10158
Balaenoptera physalus	1 emb; n~80 f	6702
	mode 1 ovulation; 2-3 rare	11619
	1.0; 0.43% twins; n>900 lt	8071
	1.01; 0.65% twins; n462 lt	9154
	1.01 emb; r1-6; 0.82% twins; n39947 f	5730
	1.02 yg; r1-4; n525 lt	8061
	1; 2.38% twins; n569 lt	1294
	1.1 cl; n20 f	3591
	2 emb ea; n3 f	9091
	3 emb; n1	856
Megaptera novaeangliae	mode 1 ovulation; 2 rare	1803
	1.01 yg; r1-2; 0.57% twins; n2979 lt	5730
	1.03; 2.68% twins; n149 lt	1802

SEXUAL MATURITY

Balaenoptera acutorostrata	f: 1st cl: 7 yr; 1960's; 13-14 yr; 1940's	5568
	f: 1st cl: 7-8 yr	8070
	f: 1st cl: 7.3 m	7372
	f: 1st cl: 7.3 m	1825
	f: 1st cl: 7.6 m	6218, 6219
	f: 3 yr; 7.5 m	9778
	f: 6 yr; 50% 8.6 m; r7.9-9.4	8069
	f: 10.5 yr; 1945; 7.7 yr; 1973; 7.98 m	869
	f: 7.2 m	8114
	m: spermatogenesis: 7-8 yr	8070
	m: 6.6-6.9 m	8114
	m: 7.47 m	869
	m: 50% 7.7 m; r6.9-8.2	8069
	m: 8.1-8.9 m	1007
	f/m: 50% 6.2 yr	5157
	f/m: 7.5 m	1375
	f/m: 8 m	5181
Balaenoptera borealis	f: 1st cl: 12.2-13.4 m	5439
	f: 1st cl: 12.9-13.2 m	6539
	f: 1st cl: 18 mo	6960
	f: 1st mate: 2 yr	6960
	f: 10-11 yr; 13.9 m	3593
	f: 1212 cm; 5% fiducial limits 1197-1257	7933
	f: 12.5-13.7 m	8110
	f: 13.3 m	9092
	f: 13.7 m	9091
	m: testes large: 12.8-13.1 m	5439
	m: testes large: > 13.5 m	3589
	m: 10-11 yr; 13.5 m	3593
	m: 1212 cm; 5% fiducial limits 1194-1227	7933
	m: 12.2-13.1 m	8110
	m: 12.6-12.9 m	6539
	m: 12.8 m	9091, 9092
	f/m: 1st mate: 2.5 yr; 13.5 m	9388
	f/m: 8-9 yr	2045
	f/m: 11.6 m	7403
	f/m: 13.7 m	1375

Balaenoptera edeni	f: 1st cl: 11.7 m	859
	f: 1st cl: 12.2 m	8067
	f: 75% 1st cl: 12.5 m; 95%ci 12.0-12.8	7933
	f: ~5 yr	10158
	m: testes large: 11.9-12.2 m; inshore 12.2-13.1 m; offshore	859, 866
	m: testes large: <12.2 m	8067
	m: ~5 yr	10158
	m: 75% sperm: 12.2 m; 95%ci 12.1-12.5	7933
	m: 12.6-13.2 m	1007
Balaenoptera musculus	f: 1st cl: ~2 yr	6252
	f: 1st cl: 22.2 m	7930
	f: 1st cl: 22.8 m	7404
	f: 1st cl: 23.7 m	6700, 6702
	f: 1st mate: 3-7 yr; r19.5-25.2 m	9389
	f: 1st birth: 23.4 m	1294
	f: ~5 yr	10158
	f: 20.4 m	9091
	f: 22.5 m	5498
	f: 27 m; from graph	1375
	m: testes large: 2-7 yr, 22.3 m	9389
	m: testes large: 21.6 m	7404
	m: testes large: 22.6 m	6700, 6702
	m: testes increase: 23.4 m	7930
	m: ~5 yr	10158
	m: 22.35 m; r21.3-23.4; n2	5498
	m: 25.5 m; from graph	1375
	f/m: increase anterior pituitary size: 22.5 m	7937
	f/m: 2nd winter	5630
	f/m: 8-10 yr	2045
	f/m: 18.9 m	1375
Balaenoptera physalus	f: increase anterior pituitary wt: 20.1 m	7937
	f: 1st cl: 6 yr, 18.3 m	8071
	f: 1st cl: 6-7 yr	7402
	f: 1st cl: 6-7 yr, 20 m	6534
	f: 1st cl: 17.1-17.4 m	4498
	f: 1st cl: 19.5 m	7404
	f: 1st cl: 19.9 m, 3 yr?	6700
	f: 1st cl: 20.1 m	7930
	f: 1st preg: 18.3 m	4147
	f: 1st birth: 19.5-19.8 m	1294
	f: 6-8 yr	11192
	f: 17.1-19.5 m	8110
	f: 17.4 m; 12 yr; 1954; 6 yr; 1975	8068
	f: 18.0 m; r18.3-18.6	4498
	f: 18.6 m	5498
	f: 18.6 m	8071
	f: 18.6 m	9091
	f: mode 19.2 m; r16.5-21.3	6875
	m: increase anterior pituitary wt: 18.6 m	7937
	m: testes large: 5-13 yr varies	3591
	m: testes large: 6 yr; 17 m	8071
	m: testes large: 17.1-17.4 m	4498
	m: testes large: 18.3 m	7404
	m: testes large: 19.2 m	6700
	m: testes large: 19.2 m; 6-7 yr	6534
	m: testes increase: 18.9 m	7930
	m: 17.5 m	9091
	m: 17.7 m	8071
	m: 17.7-18.0 m	8110
	m: 18.0-20.1 m	5498
	m: 18.3 m; 11 yr; 1954; 4 yr; 1975	8068
	f/m: 1st mate: 3 yr, 18.0 m	9388
	f/m: min 4 yr; 6 yr	8071

	f/m: 50% ~7 yr; 6-7 growth layers ear plugs	6538
	f/m: 8 yr; 1955-1979	6537
	f/m: 10.5 yr; 1941-1950	6537
	f/m: 11.5-12 yr; 1900-1928	6537
	f/m: 15.2 m	7403
	f/m: 19.5 m	1375
	f/m: est 21 m	8063
Megaptera novaeangliae	f: 1st cl: 22 mo	6958
	f: 1st cl: 5-6 yr, 11.7-12.0 m	7928
	f: 1st cl: 11.4 m	7404
	f: 1st cl: 11.4 m (33% mature); 12.3 m (100%)	8111
	f: 1st ovulate: 11.55 m; ±0.05; n29	1801
	f: 1st preg: 13.7 m	4147
	f: 5.0 yg; r4-6; n2 f	1846
	f: 11.4 m	5498
	f: 12 m	9091
	m: spermatogenesis: est 3 yr	3038
	m: spermatogenesis: 11-12 m	6958
	m: testes large: 5 yr; 11.1-11.7 m	7928
	m: testes large: 11 m; r10-12.25	1800
	m: testes large: 11.4 m (75% mature); 12.0 m (100%)	8111
	m: 11 m	10158
	m: 11.7 m	9091
	f/m: 11.9 m	1375
GESTATION		
Balaenoptera acutorostrata	9.5-10 mo; from size of newborn	5182
	10 mo	4147
	10 mo	5157
	10 mo	8114
	~10 mo	10394
	10-11 mo	5181
	10-11 mo	6952
Balaenoptera borealis	324 d; fetal growth data	6539
	<11 mo	7403
	12 mo	3589, 3593
	~12 mo	6960
Balaenoptera edeni	12.7 mo; fetal growth data	9092
Balaenoptera musculus	9 mo; fetal growth data	3488
	10-12 mo	6700, 6702
	10.75 mo; fetal growth data	6279
	>1 yr	4147
Balaenoptera physalus	9-10 mo; fetal growth data	3488
	10-11 mo	6701
	11 mo; fetal growth data	7713
	11.25 mo	6281
	11.25 mo; fetal growth data	6279
	est 11.5 mo; fetal growth data	6702
	12 mo 10 d	7403
	~1 yr; field data	4498
Megaptera novaeangliae	~11 mo	6958
	~11 mo; fetal growth data	4147
	11.5 mo	10158
	12 mo; fetal growth data	3488
	~12 mo	1802
LACTATION		
Balaenoptera acutorostrata	wean: ~6 mo	5157
Balaenoptera borealis	wean: ~5 mo	2045
	wean: 6 mo	3589, 3593
	wean: est 6 mo	6539
	wean: 1.5 yr	6960
Balaenoptera edeni	wean: 9 mo	9092
Balaenoptera musculus	wean: 6 mo	6702
	wean: ~7 mo	5630

Balaenoptera physalus	wean: ~6 mo	6701, 6702
	wean: ~6 mo	5630
Megaptera novaeangliae	wean: ~5 mo	6958
	wean: 10-11 mo	10158
	nurse: 1 yr	10698
	wean: 2nd winter	1845
INTERLITTER INTERVAL		
Balaenoptera acutorostrata	1.3 yr	10158
	14 mo	5157
	15.3 mo; min 14 mo	869
	18-24 mo	8114
Balaenoptera borealis	2 yr	6960
	2 yr	3589
Balaenoptera musculus	~2 yr	6702
	2-3 yr	6252
Balaenoptera physalus	~16 mo	6092
	2 yr	6701, 6702
Megaptera novaeangliae	~1 yr	6092
	2 yr; r1.5-2	6958
	2.4 yr; mode 2-3; r1-4; n28	1845
SEASONALITY		
Balaenoptera acutorostrata	mate: Jan-Mar; n Atlantic	4147
	mate: Feb-Mar or Aug-Sept; 2 popn; Pacific	8114
	mate: Apr?; NEW, Canada	7372
	mate: est June-Sept; Durban, S Africa	869
	mate: Dec-Mar; sw seas of Japan	6952
	birth: Oct-Jan; sw seas of Japan	6952
	birth: Nov-Jan; n Atlantic	4147
	birth: Dec-Jan or June-July; 2 popn; Pacific	8114
Balaenoptera borealis	testes: same yr rd; s hemisphere	3589
	mate: Jan; Japan	7403
	mate: Mar-Oct, peak July; S Africa	6960
	mate: est winter; n Atlantic	4147
	mate: yr rd, peak May; Atlantic	303
	mate/birth: winter; Antarctica	3591, 3593
	conceive: Apr-Aug; s hemisphere	3589
	conceive: end Nov-mid Mar; Antarctica	9154
	birth: May-Oct, peak July; S Africa	6960
	birth: ~July; S Africa	10158
	birth: Sept-Mar, peak Nov; ne Pacific	9091, 9092
	birth: Nov; Japan	7403
Balaenoptera edeni	conceive: est yr rd; few data; S Africa	859
	conceive: yr rd; inshore; fall; offshore; w Cape Province, S Africa	866
	birth: Sept-Mar, peak end Nov; e N Pacific	9092
Balaenoptera musculus	testes: large Apr-June; S Georgia, Antarctica	6700
	estrus: peak winter; S Africa	10158
	ovulate: June-Aug; S Africa, Antarctica	6702
	mate: Apr; n hemisphere	3488
	mate: Apr-Aug, peak end May-early June; S Africa	3592
	mate: June-Aug; s hemisphere	3488
	mate: yr rd, peak June-Aug; Antarctica	9154
	mate/birth: winter; Antarctica	3591
	mate/birth: yr rd; n Atlantic	4147
	birth: Mar-June; S Africa, Antarctica	6702
	birth: peak mid Apr; S Africa	3592
Balaenoptera physalus	spermatogenesis: Apr-June; S Georgia, Antarctica	6700
	estrus: 21% summer; Iceland	6538
	mate: Jan; Japan	7403
	mate: Mar-Apr; n hemisphere	3488
	mate: Apr-Aug, peak June-July; s hemisphere	7713
	mate: May; Antarctica	6701
	mate: July-Sept; s hemisphere	3488
	mate: Dec; Iceland	6538

	mate: yr rd, most Nov-Jan, peak Dec; n Pacific	8071
	mate/birth: Jan; Japan	7403
	mate/birth: Jan-Mar; n Atlantic	4147
	ovulate: peak June-Sept; s hemisphere	11192
	ovulate/sperm in testes: May-Aug; S Africa, Antarctica	6702
	conceive: end Jan-early Feb; Norway	4498
	conceive: June; Antarctica	6701
	conceive: June-Oct 70%; Antarctica	9154
	conceive: mid Nov-early Apr 68%; Norwegian Sea	9154
	conceive: yr rd, peak Apr-Aug, low Feb; Antarctica	6281
	conceive/birth: Apr-Aug; Antarctica	3591
	birth: peak early Jan; Norway	4498
	birth: Apr-Aug, peak June-July; S Africa, Antarctica	6702
	repro: Sept-Mar, peak Nov; ne Pacific	9091
Megaptera novaeangliae	testes: large yr rd; w Australia	1800
	CL present: June-Nov, peak July; w Australia	1799, 1803
	mate: Apr-May; n Atlantic	4147
	mate: Apr-May; n hemisphere	3488
	mate: June-Nov, peak Aug; Australia	10526
	mate: July-Sept; s hemisphere	3488
	mate: Aug-Nov, peak Sept; S Africa	6958
	mate/birth: winter; Australia	10698
	birth: July-Oct, peak Aug; w Australia	1802
	repro: Oct-Apr, peak Dec; ne Pacific	9091

Order Cetacea, Family Balaenidae

Right whales (*Balaena glacialis, B. mysticetus, Caperea marginata*) inhabit temperate and arctic waters. These 23,000-67,000 kg whales subsist on krill filtered through baleen. Females have 2 teats (2166, 9695). Details of anatomy (6959, 9695), copulation (1454, 2683, 3125), and milk composition (11980) are available.

NEONATAL SIZE		
Balaena glacialis	500-600 cm	8066
	600 cm	6700
Balaena mysticetus	304-427 cm	9695
	~400-405 cm	7846
Caperea marginata	est 190 cm; r160-220	9302
WEANING SIZE		
Caperea marginata	est 3.5 m; r3.2-3.8	9302
LITTER SIZE		
Balaena glacialis	1	2684
Balaena mysticetus	1; 2 rare	9695
Caperea marginata	1 emb; n1 f	4131
	1 yg; n1 f	7099
	1 emb ea; n2 f	7613
	1 ea; n2 lt	7370
SEXUAL MATURITY		
Balaena glacialis	f: 1st ovulate: 14.5-15.5 m	8113
	f: 1st birth: est 11 yr	3625a
	f: ~10 yr, 15.5 m	2166
	m: spermatogenesis: 15-16 m	8113
	m: ~10 yr, 15 m	2166
Balaena mysticetus	m: 11.58 m	6830
Caperea marginata	f: smallest preg: 6 m	510
GESTATION		
Balaena glacialis	9-10 mo	2684
	est 11-12 mo	865
	est 13-14 mo	3625a
Balaena mysticetus	~12 mo	6830
	est 13-14 mo	7846

LACTATION		
Balaena mysticetus	wean: est 1 yr	7846
INTERLITTER INTERVAL		
Balaena glacialis	2 yr	2684
	min 2 yr	865
	est 3 yr; r2-5	3625a
SEASONALITY		
Balaena glacialis	mate: Aug-Sept; Algoa Bay, S Africa	2683, 2684
	mate: winter; n Pacific, n Atlantic	2166
	preg f: June-July; n Atlantic	1980
	birth: June-July; S Africa	2684
	birth: peak Aug; s,w S Africa	868
	birth: Dec-Mar; n Atlantic	3625a
	birth: spring; n Pacific, n Atlantic	2166
Balaena mysticetus	mate: May; n1; n AK	3125
	mate: end summer; Greenland	9695
	birth: Feb-Mar; Greenland	9695
	birth: Mar-Aug, peak May; AK	6830, 7846
Caperea marginata	preg f: June, Sept, Nov-Dec; n3; s hemisphere	510